Nirman Achintha Wedikkara

Information Technology Undergraduate | Aspiring DevOps, Cloud, AI & SE Intern

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Professional Summary

Performance-driven Information Technology Undergraduate seeking an internship focused on DevOps, Cloud, AI, and Software Engineering. I bring hands-on expertise in automating delivery pipelines, managing Linux servers, and containerizing workloads. As the Founder of Vibe Solution, I architect secure cloud infrastructure, deploy microservices, and orchestrate server environments. I integrate AI models and maintain strong full-stack software engineering skills to build highly available systems.

Technical Skills & Competencies

- **Cloud, DevOps & Systems:** Docker, Docker Compose, Linux Server Administration (Ubuntu, Debian), Nginx (Reverse Proxy & SSL/TLS), GitHub Actions (CI/CD), AWS, Microsoft Azure (AZ-900 in progress), Bash Shell Scripting
- **AI & Advanced Systems:** Hugging Face Spaces, Transformer Models (LLMs), NLP Pipelines, Model Context Protocol (MCP), Prompt Engineering
- **Databases & Architecture:** PostgreSQL (Schema Migrations & Indexing), MySQL (3NF Relational Modeling), Microservices, Clean Architecture
- **Programming Languages:** Python, JavaScript (ES6+), TypeScript, Dart, Java, PHP, Go (Golang), SQL, HTML5, CSS3/Tailwind CSS
- **Web & Mobile Engineering:** React.js, Next.js, Node.js, Express.js, Flutter (Android/iOS), Vue.js, RESTful APIs, Supabase, SQLite, State Management
- **Soft Skills:** Technical Leadership, Analytical Problem Solving, Team Collaboration, Effective Communication, Strategic Planning

Featured Engineering Projects

FLAGSHIP INNOVATION ASVANNA - Agricultural Market Intelligence & Zero-Waste Platform (2025-2026)

Lead Architect & Full-Stack Systems Developer | ITUM University of Moratuwa | github.com/NirMAN-15/asvanna

- **Multi-Container Orchestration & DevOps:** Containerized PostgreSQL, Express REST API, and React frontend using Docker Compose. Configured GitHub Actions automated CI pipelines and production deployment on a Linux cloud VPS behind an Nginx reverse proxy with SSL encryption.
- **National Agritech System:** Spearheaded the end-to-end design and engineering of an agritech ecosystem developed for the Division of Information Technology, ITUM (University of Moratuwa), resolving harvest price volatility and intermediate broker monopolies.
- **Predictive Saturation Risk Engine:** Engineered a pre-cultivation saturation calculation engine with a 3-tier risk notification model (Safe < 70%, Warning 70-85%, Over-Planted > 85%) featuring intelligent multi-factor alternative crop suggestions based on regional land data.
- **Geo-Fenced Surplus Marketplace:** Architected a real-time localized trading platform (5 km radius) enabling direct transactions between local farmers and commercial buyers (wholesalers, supermarkets, hospitality) to eliminate post-harvest agricultural waste.
- **Tech Stack:** Node.js, Express.js, PostgreSQL 15, React 18 (Vite), Flutter 3.x, Docker Compose, GitHub Actions, Nginx, Linux VPS.

Kapruka AI Shopping Assistant - Conversational E-Commerce Platform (2026)

AI & Web Systems Engineer | React.js, Express.js (Serverless), Model Context Protocol (MCP), Vercel

- Connected a React frontend to Express serverless proxies communicating with a remote Kapruka Model Context Protocol (MCP) server for live product catalog querying and intent clarification.
- Developed an intelligent multilingual conversational shopping assistant supporting English, Sinhala, and Tanglish for Sri Lanka's largest e-commerce platform (Kapruka.com).

AI Learning Assistant & Educational NLP Engine (2025-2026)

AI Developer | Python, Hugging Face Spaces, Transformers

- Deployed an educational AI assistant on Hugging Face Spaces utilizing open-source NLP transformer pipelines to provide structured coding solutions and CS tutorials.
- Integrated prompt workflows and context-aware responses to streamline CS conceptual learning and student doubt resolution.

Distribution Management System (DMS) - Offline-First Enterprise Suite (2025-2026)

Full-Stack & Mobile Developer | Flutter, Node.js, Express.js, Supabase (PostgreSQL), SQLite, Bluetooth ESC/POS

- **Offline-First Field Architecture:** Engineered a resilient cross-platform mobile application in Flutter for field sales representatives, incorporating SQLite local databases for order processing in remote offline zones.
- **Two-Way Synchronization:** Built automated "Morning Sync" and background cloud delta sync routines connecting local mobile storage to a cloud-hosted Supabase PostgreSQL backend.
- **Hardware Integration:** Implemented direct ESC/POS wireless printing via Bluetooth to 58mm portable thermal printers, generating instantaneous customer invoices and receipts on site.

Enterprise ERP Asset Management Module (2025-2026)

Software Developer | Go (Golang), Vue.js

- Built full-stack asset tracking modules in Go and Vue.js for managing company hardware, status transitions, disposal records, and searchable asset inventories.
- Implemented performant backend services and audit trails to ensure enterprise data consistency and hardware tracking accountability.

Career Path (Career Compass) - Diagnostic IT Career Analytics Engine (2025)

Lead Full-Stack Architect | PHP, MySQL, JavaScript, HTML5/CSS3, Linux Hosting | Live: careerpath.fwh.is

- Designed and normalized a 3NF relational MySQL schema with automated backups, deploying to Linux cloud hosting and successfully handling high student concurrency during national university launches.

Professional Experience

Vibe Solution January 2025 - Present
Founder & Lead Systems Developer | Global / Remote (vibesolution.tech)

- Configured and maintained Linux cloud production environments (Ubuntu/Debian), managing DNS routing, SSL/TLS certificates, Nginx reverse proxy configurations, and Docker staging environments.
- Integrated automated LLM workflows and conversational customer bots, cutting client inquiry response times by over 40%.
- Engineered custom full-stack solutions with React, Next.js, Node.js, and Python, consistently maintaining sub-second render performance and 99.9% uptime across production deployments.

Bandarawela Sussex & Seevali Schools January 2024 - January 2025
Strategic Chess Coach & Analytical Instructor | Bandarawela, Sri Lanka

- Engineered systematic tactical training frameworks that boosted student qualification rates by 25% at provincial championships.

Education & Academic Qualifications

National Diploma in Technology (NDT) in Information Technology Expected 2026
Institute of Technology, University of Moratuwa (ITUM)

- **Relevant Coursework:** Data Structures & Algorithms, Database Management Systems (DBMS), Object-Oriented Programming (OOP), Computer Networks, Operating Systems, Software Engineering, Cloud Computing, Web & Mobile Development.

G.C.E. Advanced Level - Physical Science (Mathematics) Stream Concluded 2023
Bandarawela Central College | Combined Mathematics (S), Physics (C), Chemistry (C)

Licenses & Professional Certifications

- **AWS Certified Cloud Practitioner (CLF-C02):** Learning path completed (Official Certification in progress)
- **Microsoft Certified: Azure Fundamentals (AZ-900):** Microsoft Learn Curriculum completed (Official Certification in progress)
- **Python for Beginners:** Center for Open & Distance Learning, University of Moratuwa (Credential ID: oKphzwlioe | Issued Nov 2024)
- **Certified National Chess Instructor (Level 1):** Chess Federation of Sri Lanka (CFSL | Issued 2023)

Community Leadership, Honors & Extracurricular Service

- **ZeroPlastic Movement & NDT Club (Voices Beyond Plastic 2026/27):** Awarded Official Letters of Appreciation as Organizing Committee Member for environmental sustainability initiatives and as Street Drama Crew Member for public education performances.
- **Certificate of Participation - ZeroPlastic TRAIL 26:** Recognized by Nishshanka de Silva for contributions to nationwide cleanup and sustainability initiatives.
- **School Board of Prefects (2019-2022):** Bandarawela Central College. Coordinated student leadership, institutional discipline, and large-scale cultural and academic functions across 1,000+ students.
- **Cadet Air Wing Boys Annual Assessment Camp (2018-2021):** Completed intensive survival, aerospace fundamentals, operational drills, and military-grade leadership training.
- **Uva Provincial Chess Championship Team Member (2014-2022):** Represented Bandarawela Central College across provincial and national tournaments, earning individual and team accolades.

Professional References

Mrs. Uthpala Athukorala
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